

The Networks of Support Approach

Essential factors for effective prevention programming

Evidence

1 - Why Prevention?:

Effectiveness Trial of Wingman-Connect Implemented Across Career Phases

Submitted to the National Institute of Mental Health, PAR21-130, 10/13/2022

???? Possibilities: Some data but no paper, yet - quote not publicly accessible:

<https://classic.clinicaltrials.gov/ct2/show/NCT05967364>

<https://ctv.veeva.com/study/career-enhancement-training-study-delivered-across-career-phases>

Upstream prevention is necessary.

Existing military suicide intervention programming focuses on detecting and remediating risk after suicidal individuals are identified. Although undoubtedly saving lives, this approach is insufficient on its own.

Wingman-Connect addresses an urgent need for preventions that are effective in reducing suicide risk across broad populations – particularly where stigma and concerns about negative career impacts remain major barriers to disclosing suicidal thoughts/behaviors.

[Peer-adult network structure and suicide attempts in 38 high schools: implications for network-informed suicide prevention \(10,291 students\)](#)

Journal of Child Psychology and Psychiatry *: * (2019), Peter A. Wyman, Trevor A. Pickering, Anthony R. Pisani, et al.

To date, suicide prevention has focused primarily on individual-level psychiatric risk factors and on strategies to refer and treat high-risk youth. Yet researchers and policymakers recommend strengthening positive social bonds because social integration can reduce suicidal behavior in youth

To our knowledge, this is the first study of any adolescent health problem, including suicidal behavior, that integrates adult connections into friendship networks at a school population level.

Network interventions addressing these processes, such as maximizing youth–adult connections schoolwide and heightening influence of youth with healthy coping, could create more protective schools. The present study is the first empirical evidence linking more integrated youth–adult networks (i.e., students sharing trusted adults with friends) to lower Suicide Attempts.

[Effect of the Wingman-Connect Upstream Suicide Prevention Program for Air Force Personnel in Training. A Cluster Randomized Clinical Trial](#)

JAMA Open Network, October, 2020

Group training that builds cohesive, healthy military units is promising for upstream suicide prevention.

The Wingman-Connect program used group skill building for cohesion, shared purpose, and managing career and personal stressors. Wingman-Connect is the first universal prevention program to reduce suicidal ideation and depression symptoms in a general Air Force population. Group training that builds cohesive, healthy military units is promising for upstream suicide prevention.

2 - Why Ecological Validity?:

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Data-informed implementation preparation.

A “voltage drop” and poor sustainment are common as interventions transition from controlled efficacy trials to regular use.

Conversely, scaled out interventions are more impactful with careful preparation using implementation science frameworks.

3 - Why Evidence-based Methodology?:

[Effect of the Wingman-Connect Upstream Suicide Prevention Program for Air Force Personnel in Training. A Cluster Randomized Clinical Trial](#)

JAMA Open Network, October, 2020

Research Outcomes – Evidence Based Key Findings include:

- Increased class cohesion, morale, positive bonds, and acceptability of help seeking
- Reduction in suicide risks scores, depression, and reactivity to anger
- Nearly 50% reduction in occupational impairment (corrective training, negative counseling)

Measurable Impact on both Protective Factors and Risk Factors

Wingman-Connect was tested using the **gold-standard research design with Airmen in Training** from 2017-2019 (randomized controlled trial). Training showed statistically significant increases in group and individual level protective factors, as well as reductions in negative outcomes including and depression and suicide risk.

Social Science & Medicine

[Social Science & Medicine 296 \(2022\) 114737](#)

[**Wingman-Connect Program increases social integration for Air Force personnel at elevated suicide risk: Social network analysis of a cluster RCT**](#)

The current attention on the detection and treatment of suicidal military members is necessary, but ultimately insufficient to meet the needs of many suicidal service members. Results from our study suggest that the status quo progression for many service members at elevated suicide risk is one of worsening disconnection, evidenced by the trajectory of declining valued connections found in the active control condition.

4 - Why Consistency?:

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JAMA Open Network, October, 2020

Wingman-Connect's group training design ecologically-valid and appealing for participants.
Wingman-Connect development benefited from strong organizational collaboration, resulting in an ecologically valid program.

5 - Why Discovery?:

Discovery of personal strengths and how those strengths impact our social network, and the discovery of other's strengths and how those strengths impact us and the social network. Everyone's competencies are recognized.

[Dr. Peter Wyman's interview 2022](#)

6 - Why Whole Community?:

Everyone is invited and contributes to the overall health of the social network.

[Social Science & Medicine](#) [Social Science & Medicine 296 \(2022\) 114737](#)

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Findings support a growing case for the unique contribution of group-level interventions to improve social health of broader populations.

The Wingman-Connect Program trains all members of natural organizational units together to strengthen group bonds and the diffusion of suicide-protective norms and practices. Skill-building activities focus on relationships and practices supportive of mental health.

Group skill-building activities identify strengths of all group members, and members learn how a strong network supports all members' health and fitness. W-CP uses high energy activities and peer-to-peer teaching; exercises inside and outside of training promote adoption of skills into unit culture.

7 - Why Peer Influence?:

Social networks provide the mechanisms for the diffusion of norms and practices and the context for peer group monitoring and support. Positive social bonds augment the protective factors while learning and modeling skills for growing relationships and developing connections to each other. These connections reduce 'Pruning'.

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WINGMAN-CONNECT PROGRAM MODEL AND DEVELOPMENT

Wingman-Connect is an adaptation of a network-informed intervention. Since

2010, PI Wyman has led research (NIH, CDC funded) testing that program, which disseminates skills for social health through youth peer networks. The high-energy, interactive training improves student connectedness and coping norms, and protective effects spread school-wide including adult help for suicidal youth. A study aggregating three RCTs (N=78 schools; 39,900 students) showed fewer suicide deaths in schools implementing this approach.

Wingman-Connect Program structure and key elements together strengthen relationship networks, cohesion, adaptive coping and group norms.

[Effect of the Wingman-Connect Upstream Suicide Prevention Program for Air Force Personnel in Training. A Cluster Randomized Clinical Trial](#)

JAMA Open Network, October, 2020

Airmen learn skills to grow and sustain protective factors essential to job success, mental health and reduced suicide risk (Kinship, Purpose, Guidance, Balance). Class exercises create more cohesive units with skills extended into group culture. Wingman Connect training is delivered to organizational units using interactive exercises to build key group and individual protective factors (Four Cores):

Social Science & Medicine

[Social Science & Medicine 296 \(2022\) 114737](#)

[Wingman-Connect Program increases social integration for Air Force personnel at elevated suicide risk: Social network analysis of a cluster RCT](#)

Wingman-Connect counteracted the typical **drift** towards disconnection for Airmen at elevated suicide risk. Six-month findings again showed that W-CP offset the typical trajectory of decreasing connections for Airmen at elevated suicide risk.

These findings are, to our knowledge, the first to show that a suicide prevention program for small personnel units improved significantly the group relationship network and socially integrated members over time.

This study extends knowledge of the social and relationship impact of the Wingman-Connect Program as it decreased suicide risk.

Wingman-Connect Program counteracted the expected drift towards disconnection for Airmen at elevated suicide risk by targeting the unit's relationship network itself. These groups built enhanced

suicide protection into their relationship networks, with the most consistent benefits for Airmen at elevated risk of suicide and for those starting with fewest connections.

8 - Why Mentoring?:

[Effect of the Wingman-Connect Upstream Suicide Prevention Program for Air Force Personnel in Training. A Cluster Randomized Clinical Trial](#)

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- Group cohesion and belonging (Kinship)
- Informal and formal support and help-seeking (Guidance)

[Social Science & Medicine](#) [Social Science & Medicine 296 \(2022\) 114737](#)

[Wingman-Connect Program increases social integration for Air Force personnel at elevated suicide risk: Social network analysis of a cluster RCT](#)

Specific modules and activities build healthy relationships and accountability (Kinship) informal and formal help-seeking (Guidance).

[An outcome evaluation of the Sources of Strength suicide prevention program delivered by adolescent peer leaders in high schools](#)

[Am J Public Health. 2010 September; 100\(9\): 1653–1661](#)

The norms most strongly enhanced through the intervention were students' perceptions that adults in their school can provide help to suicidal students and the acceptability of seeking help from adults.

9 - Why Active Learning? :

High-energy activities maintain interest, motivation, and create shared positive emotional experiences and create norms that incentivize adaptive coping skills. Space is created to practice skills that grow healthy bonds and draw upon resources in social environments to meet goals, manage career and personal challenges, and prevent behavioral health problems.

[Effect of the Wingman-Connect Upstream Suicide Prevention Program for Air Force Personnel in Training. A Cluster Randomized Clinical Trial](#)

[JAMA Open Network, October, 2020](#)

Active Learning and Transfer of Training to Life

- Drawing out 'real-world' strengths from participants as a primary method to teach Kinship, Purpose, Guidance, & Balance.
- High energy activities maintain interest, motivation, and a shared positive emotion.
- 'Real-world' exercises inside and outside of training reinforce the application of skills in groups and in individual activities of daily life.

10 - Why Deep Conversations?:

Personally meaningful - participants are invited to share personally relevant goals and values referenced throughout. Drawing upon 'real life' strengths and conversations as the primary way to teach. Provides space for culturally relevant customs, celebrations, and traditions.

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[JAMA Open Network, October, 2020](#)

Wingman-Connect impact promoting more cohesive, healthy training classes was a mechanism that reduced risk for suicidal thoughts and behaviors. More cohesive, healthy classes 'lifted up' individuals who were vulnerable to fitness and mental health problems. **Study findings suggest that group training for organizational units may be critical for 'transfer' of training concepts and skills into the organization's culture.**

11 - Why Group Cohesion?:

[Social Science & Medicine](#)

[Social Science & Medicine 296 \(2022\) 114737](#)

[Wingman-Connect Program increases social integration for Air Force personnel at elevated suicide risk: Social network analysis of a cluster RCT](#)

In more connected, healthier units, vulnerable members can borrow strength from adaptive members. Stronger units also promote job fitness, as shown by W-CP's impact reducing occupational problems, an effect that may reduce future problems that precipitate new instances of suicidal thoughts and behaviors. In that sense, W-CP may specifically assist vulnerable Airmen in overcoming barriers to forming meaningful social connections.

Wingman-Connect Program's interactive training created more dynamic relationship networks. Airmen in W-CP groups made more changes in who they named as valued connections, even as the overall average number of valued connections increased. The largest dynamic changes were for Airmen with elevated suicide risk in W-CP groups, who made on average three times more new valued connection nominations.

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[Research Outcomes – Evidence Based Key Findings include:](#)

- Increased class cohesion, morale, positive bonds, and acceptability of help seeking

Wingman Connect strengthens skills to grow and sustain healthy social bonds. Natural organization (tech school classes) units train together to incorporate skills into culture, build cohesion and shared purpose. **The objective is preventing occupational and behavioral health problems.**

The study's findings validate the underlying network health model: stronger bonds within a more cohesive healthy class reduced suicidal ideation and depression symptoms. These findings suggest that Wingman-Connect classes became increasingly unified around healthy norms and encouraged classmates who were vulnerable to mental health or occupational problems at a key juncture of military training, in addition to meeting their needs for belonging. Work with recent social network modeling show that cohesive groups serve a protective regulatory function through norms and pressures to conform.

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Novel Intervention Targeting Peer Social Networks of Military Members

The Wingman-Connect Program's novel group intervention strengthens *social bonds* as well as *peer network norms* that incentivize healthy coping. This interactive training progresses from individual engagement (e.g., each participant shares career goals and values) to building group level skills and norms (e.g., mapping group strengths).

12 - Why Connection to Guidance?:

Informal and formal help-seeking. Cohesive youth–adult networks may promote more help-seeking for students and for their friends with whom they share a common bond to a supportive adult.

[Peer-adult network structure and suicide attempts in 38 high schools: implications for network-informed suicide prevention \(10,291 students\)](#)

[Journal of Child Psychology and Psychiatry * \(2019\), Peter A. Wyman, Trevor A. Pickering, Anthony R. Pisani, et al.](#)

Social networks provide the mechanisms for the diffusion of norms and practices and the context for peer group monitoring and support. Network characteristics influence the spread of many health behaviors. By focusing on patterns of relationships, network methods can clarify the contexts in which relationships form and exert influence on others.

Cohesive youth–adult networks may promote more help-seeking for students and for their friends with whom they share a common bond to a supportive adult.

Maximizing protective bonds across school populations, increasing opportunities for group cohesion including integrated youth– adult networks, and promoting influence of youth with healthy coping. Our findings suggest effective interventions will involve youth and adults.

13 - Why Belonging?:

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14 - Why Help Seeking?:

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